

# Homework 06

## REAL VARIABLES I

### Fall 2026

**Due:** 2026-10-04 19:00

**Topics:** The non-negative integral, monotone convergence, Fatou's lemma, and measures defined by densities

#### Instructions

Submission is through Gradescope, accessible through [Blackboard](#).

**Grading rubric:** Every problem/subproblem is graded according to this rubric (no intermediate scores):

- **0%:** No submission or no meaningful content.
- **30%:** Attempt at solution, outline of idea.
- **60%:** Serious attempt at solution. Solution fails to address important points.
- **80%:** Solutions with fixable mistakes or incompleteness.
- **100%:** Complete and correct solution.

You may receive up to a 10% penalty for very unclear or messy exposition.

Please start every solution on a separate page. Clearly mark the pages associated to every part. You may submit either computer typeset solutions or handwritten solutions. Handwritten solutions must be *very clearly* legible.

You are encouraged to work in groups. You may submit the entire homework assignment as a group (up to 4 people). If you only did some exercises as a group, instead of a solution you may clearly indicate whose submission I should look at for the solution to the problem.

Indicate any external help/notes/resources you used. You may not use generative AI in any way to help you with the HW.

#### 1. (When monotonicity of the integral is strict)

Let  $(X, \mathcal{A}, \mu)$  be a measure space, and let  $f, g: X \rightarrow [0, +\infty]$  be measurable. Assume that  $g \leq f$  almost everywhere and that

$$\int_X f \, d\mu < +\infty.$$

##### 1.a (weight: 1.0) Prove the equivalence

$$\int_X f \, d\mu > \int_X g \, d\mu \iff \mu(\{x \in X : f(x) > g(x)\}) > 0.$$

**Free hint.** Since  $\int_X f \, d\mu < +\infty$ , the function  $f$  is finite almost everywhere. After discarding the null set on which  $f = +\infty$  or  $g > f$ , set  $u = f - g$  and use

$$\{u > 0\} = \bigcup_{m=1}^{\infty} \{u \geq 2^{-m}\}.$$

#### 2. (Liminf, limsup, and Fatou's lemma)

Let  $(f_n: X \rightarrow [0, +\infty])_{n \in \mathbb{N}}$  be measurable.

Recall the pointwise definitions

$$\liminf_{n \rightarrow \infty} f_n = \sup_{N \in \mathbb{N}} \inf_{n \geq N} f_n, \quad \limsup_{n \rightarrow \infty} f_n = \inf_{N \in \mathbb{N}} \sup_{n \geq N} f_n.$$

For a fixed  $x \in X$ , explain to yourself, in words or with pictures, what each expression records about the tail values  $(f_n(x))_{n \in \mathbb{N}}$ .

Make sure you understand that

$$\liminf_{n \rightarrow \infty} f_n(x) \leq \limsup_{n \rightarrow \infty} f_n(x),$$

and that equality holds if and only if  $(f_n(x))$  converges in  $[0, +\infty]$ .

**2.a** (weight: 0.7) **IN CLASS** Use the monotone convergence theorem and the preceding part to prove Fatou's lemma:

$$\int_X \liminf_{n \rightarrow \infty} f_n \, d\mu \leq \liminf_{n \rightarrow \infty} \int_X f_n \, d\mu.$$

Every step involving an infimum, supremum, or limit should be justified.

**Free hint.** Define

$$g_N = \inf_{n \geq N} f_n.$$

Show that  $g_N \uparrow \liminf_n f_n$ , apply the monotone convergence theorem, and use  $g_N \leq f_n$  for every  $n \geq N$ .

**2.b** (weight: 0.3) **IN CLASS** Give an example in which Fatou's inequality is strict, even though every  $f_n$  has the same finite integral.

### 3. (Measures defined by non-negative densities)

Let  $(X, \mathcal{A}, \mu)$  be a measure space, and let  $h: X \rightarrow [0, +\infty]$  be measurable. Do *not* assume that  $h \in L^1(\mu)$ . For  $E \in \mathcal{A}$ , define

$$\nu(E) = \int_E h \, d\mu.$$

**3.a** (weight: 0.4) Prove that  $\nu$  is a measure on  $(X, \mathcal{A})$ . Indicate precisely which integral limit theorems you use.

**3.b** (weight: 0.4) **IN CLASS** Prove that for every non-negative measurable  $\varphi$ ,

$$\int_X \varphi \, d\nu = \int_X \varphi h \, d\mu.$$

First prove the identity for indicator functions, then for non-negative simple functions, and finally for arbitrary non-negative measurable functions.

**3.c** (weight: 0.2) Prove that  $\nu$  is finite if and only if  $h \in L^1(\mu)$ . If  $\mu$  is  $\sigma$ -finite, prove in addition that  $\nu$  is  $\sigma$ -finite if and only if  $h < +\infty$  almost everywhere with respect to  $\mu$ .

### 4. (Capturing an integral on sets of finite measure)

Let  $(X, \mathcal{A}, \mu)$  be a measure space, and let  $f: X \rightarrow [0, +\infty]$  be measurable.

**4.a** (weight: 0.6) Assume that  $\int_X f \, d\mu < +\infty$ .

Prove that there is an increasing sequence  $(E_n)_{n \in \mathbb{N}}$  of measurable sets such that

$$\mu(E_n) < +\infty \quad \text{for every } n,$$

and

$$\int_{E_n} f \, d\mu \uparrow \int_X f \, d\mu.$$

Thus a finite integral can be captured by an increasing sequence of sets of finite measure, even when the ambient measure space is not  $\sigma$ -finite.

**Free hint.** Define

$$E_n = \{x \in X : f(x) \geq 1/n\}.$$

Use Markov's inequality to prove that every  $E_n$  has finite measure, and determine the union of the sets  $E_n$ .

**4.b** (weight: 0.4) Prove that, independently of whether  $\int_X f \, d\mu$  is finite or infinite, if  $(E_n)_{n \in \mathbb{N}}$  is a sequence of measurable sets such that

$$E_n \uparrow X,$$

then

$$\int_{E_n} f \, d\mu \uparrow \int_X f \, d\mu.$$

Then show that if  $(X, \mathcal{A}, \mu)$  is  $\sigma$ -finite, there exist measurable sets  $(E_n)_{n \in \mathbb{N}}$  such that

$$E_n \uparrow X,$$

$$\mu(E_n) < +\infty \quad \text{for every } n,$$

and

$$\int_{E_n} f \, d\mu \uparrow \int_X f \, d\mu.$$

In particular, the last limit is  $+\infty$  when  $\int_X f \, d\mu = +\infty$ .